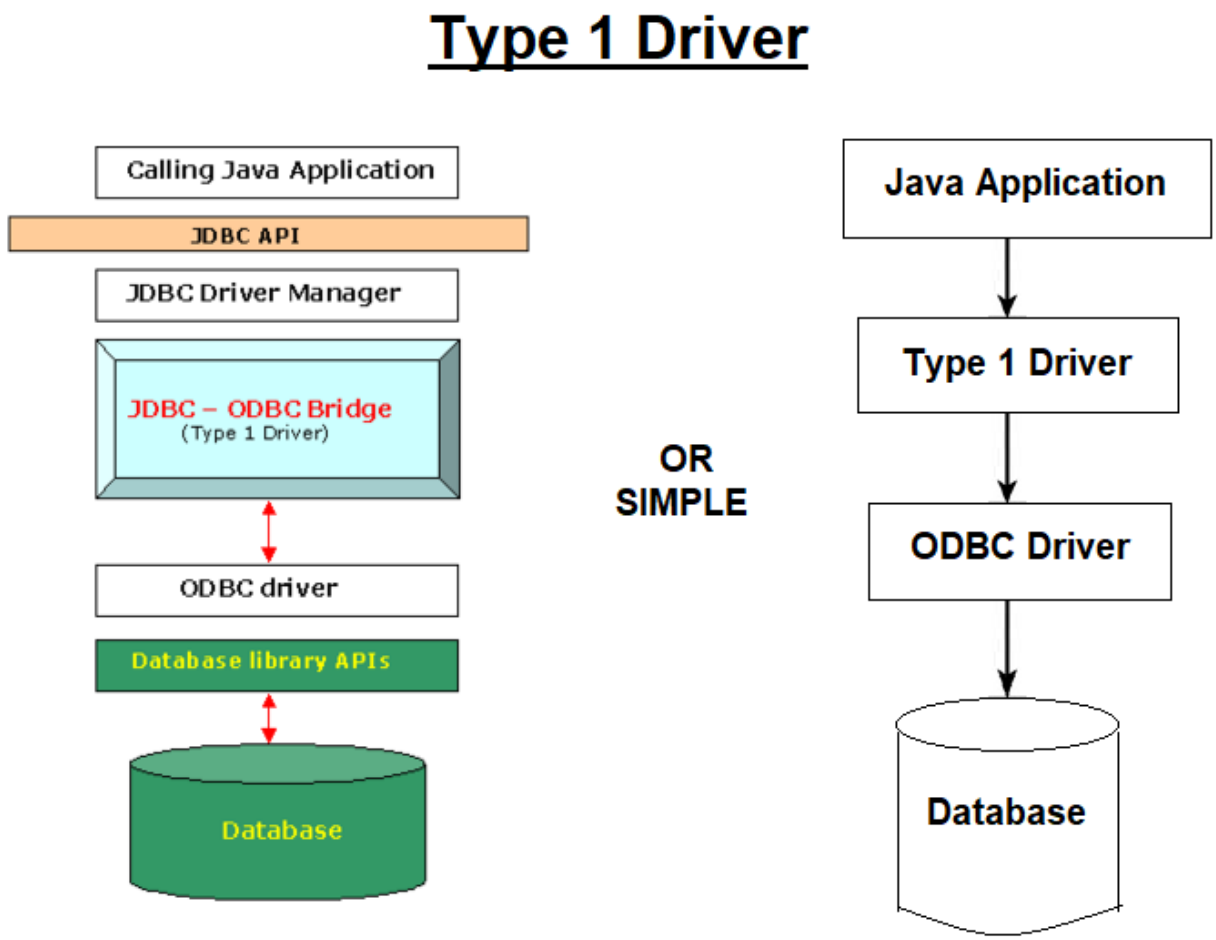
Smart Programming : YouTube Channel

An investment in Knowledge pays the best interest….

JDBC (PART 3)

Type 1 driver

Architecture Diagram:



(image taken from wikipedia)

Other names: JDBC ODBC Bridge Driver OR Bridge Driver

Architecture type: 2 Tier Architecture (because there are two parts i.e. Client Machine and Database)

Thin or thick driver: Thick driver

Description:

- Type 1 driver is provided by Sun Microsystems as the part of JDK.
- It converts JDBC calls into ODBC calls and then ODBC driver converts these ODBC calls into database specific calls and vice versa

Does it use any other driver: Yes, ODBC driver

Other driver details: ODBC was created by Microsoft and Simba Technologies. Open Database Connectivity (ODBC) is a standard application programming interface (API) for accessing database management systems (DBMS)

Database dependency: Database Independent as it does not directly communicate with database

Platform dependency: Platform Dependent because it uses ODBC which is created by Microsoft

Used for: Used for Standalone Applications

Advantages:

1. It is very easy to use and maintain.
2. We are not required to install type 1 driver as it was the part of JDK
3. Easy to migrate from one database to another as it does not directly communicate with database

Disadvantages:

1. Performance is very slow as JDBC calls is converted into ODBC calls and then ODBC calls convert into database specific calls
2. The ODBC driver needs to be installed on the client machine
3. It is platform dependent as type 1 driver internally depends on ODBC driver which will work only on windows machine
4. No support from JDK 8 (was available till JDK 1.0 to 1.7 versions)

Can be used with MySQL: Yes

Can be used with Oracle: Yes but from Java 8 Oracle does not support JDBC ODBC bridge.

Working:

Driver Class Name: sun.jdbc.odbc.JdbcOdbcDriver

JDBC URL/Connection URL: jdbc:odbc:xxxdsn

Username: root (MySQL) or system/scott (Oracle)

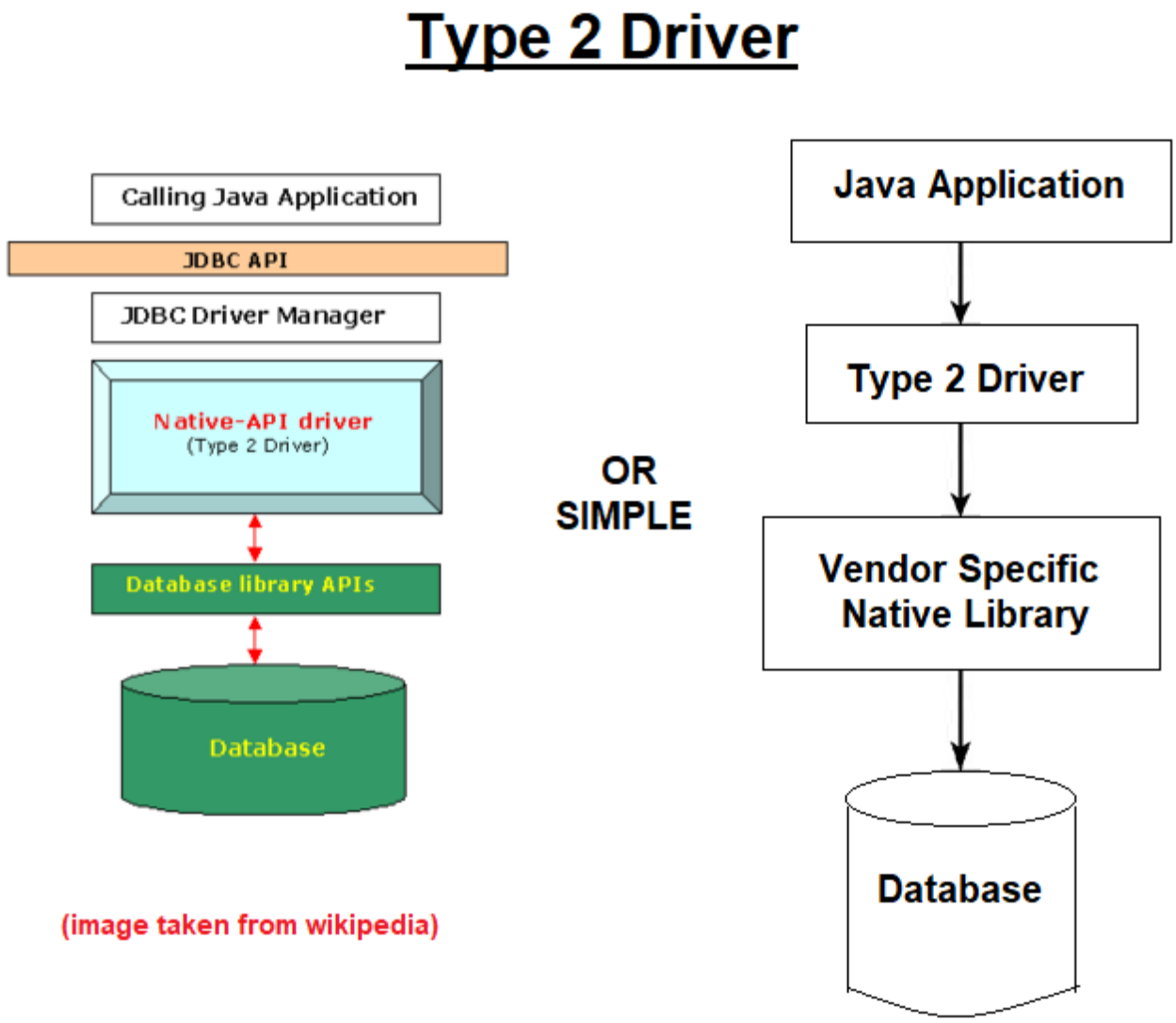
Password: root (MySQL) or system/tiger (Oracle)

Jar files: No jar file required

Smart Programming
We Educate
We Develop

Type 2 driver

Architecture Diagram:



(image taken from wikipedia)

Other names: Native API Partly Java Driver OR Native-API Driver

Architecture type: 2 Tier Architecture (because there are two parts i.e. Client Machine and Database)

Thin or thick driver: Thick driver

Description:

- Type 2 driver is approx. same as type1 driver except that ODBC driver is replaced with database vendor specific native libraries.
- In type 2 driver, we have to install native libraries on the client machine which are provided by database vendors.
- It converts JDBC calls into vendor specific native libraries calls which can be understand by database directly.

Does it use any other driver: Database specific native libraries

Other driver details: Database vendors provide these native libraries.

Native libraries means the set of functions written in non-java languages, mostly C & C++.

Database dependency: Database Dependent because it internally uses database specific native libraries

Platform dependency: Platform Dependent

Used for: Used for Standalone Applications

Advantages:

1. Its performance is better as compared to type 1 driver because it required only one level conversion from JDBC calls to native library calls
2. Portable as compared to type 1 driver because type 1 driver is applicable only for windows platform.
3. No need to arrange ODBC drivers.

Disadvantages:

1. It is database dependent driver because it internally uses database specific native libraries. Due to this it was difficult to migrate from one database to another database.
2. It is platform dependent.
3. We are required to install vendor client library on client machine.
4. Not all database provides client-side library for example Oracle provides the client-side library but MySQL don't.

Can be used with MySQL: No

Can be used with Oracle: Yes

Working:

Driver Class Name: oracle.jdbc.driver.OracleDriver OR

oracle.jdbc.OracleDriver

JDBC URL/Connection URL: jdbc:odbc:oci:@XE

Username: system/scott

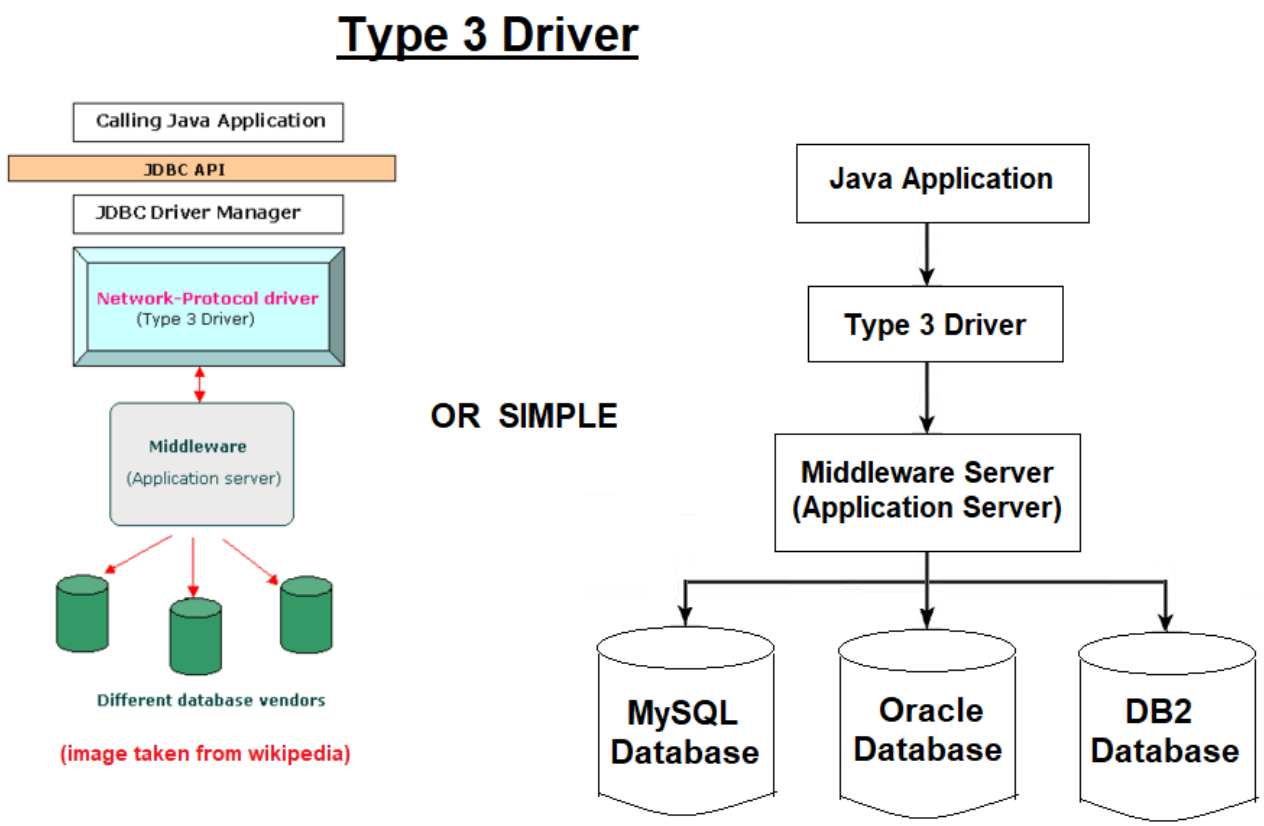
Password: system/tiger

Jar files: ojdbc14.jar OR ojdbc6.jar OR ojdbc7.jar

Smart Programming
We Educate
We Develop

Type 3 driver

Architecture Diagram:



Other names: All Java Net Protocol Driver OR Network Protocol Driver OR Middleware Driver

Architecture type: 3 Tier Architecture (because there are three parts i.e. Client Machine, Application Server (Middleware) and Database)

Thin or thick driver: Thick driver

Description:

- Type 3 driver converts JDBC calls directly or indirectly into a vendor specific database protocol
- Making use of middleware provides additional advantages of security and firewall access
- The type 3 driver is written in java
- Can be used for multiple database – Not vendor specific

Does it use any other driver: Application Server (Middleware Server)

Other driver details: If we want to use Middleware server then we can download any application server for eg. IDS Server from www.idssoftware.com website (download trial version)

Database dependency: Database Independent because it does not directly communicate with database.

Platform dependency: Platform Independent as the driver is created in java.

Used for: Used for Web Applications or Distributed Applications

Advantages:

1. Since the communication between client and the middleware server is database independent, there is no need for the database vendor library on the client. The client need not be changed for a new database.
2. A single driver can handle any number of databases, provided the middleware supports it.
3. It is platform independent and database independent.

Disadvantages:

1. Requires database-specific coding to be done in the middle tier.
2. The middle ware layer added may result in additional latency, but is typically overcome by using better middle ware services.
   (Latency is the delay between a user's action and a web application's response to that action, often referred to in networking terms as the total round trip time it takes for a data packet to travel)

Can be used with MySQL: Yes

Can be used with Oracle: Yes

Working:

Driver Class Name: ids.sql.IDSDriver

JDBC URL/Connection URL:

jdbc:ids://localhost:12/conn?dsn=xxxdsn

Username: root (MySQL) or system/scott (Oracle)

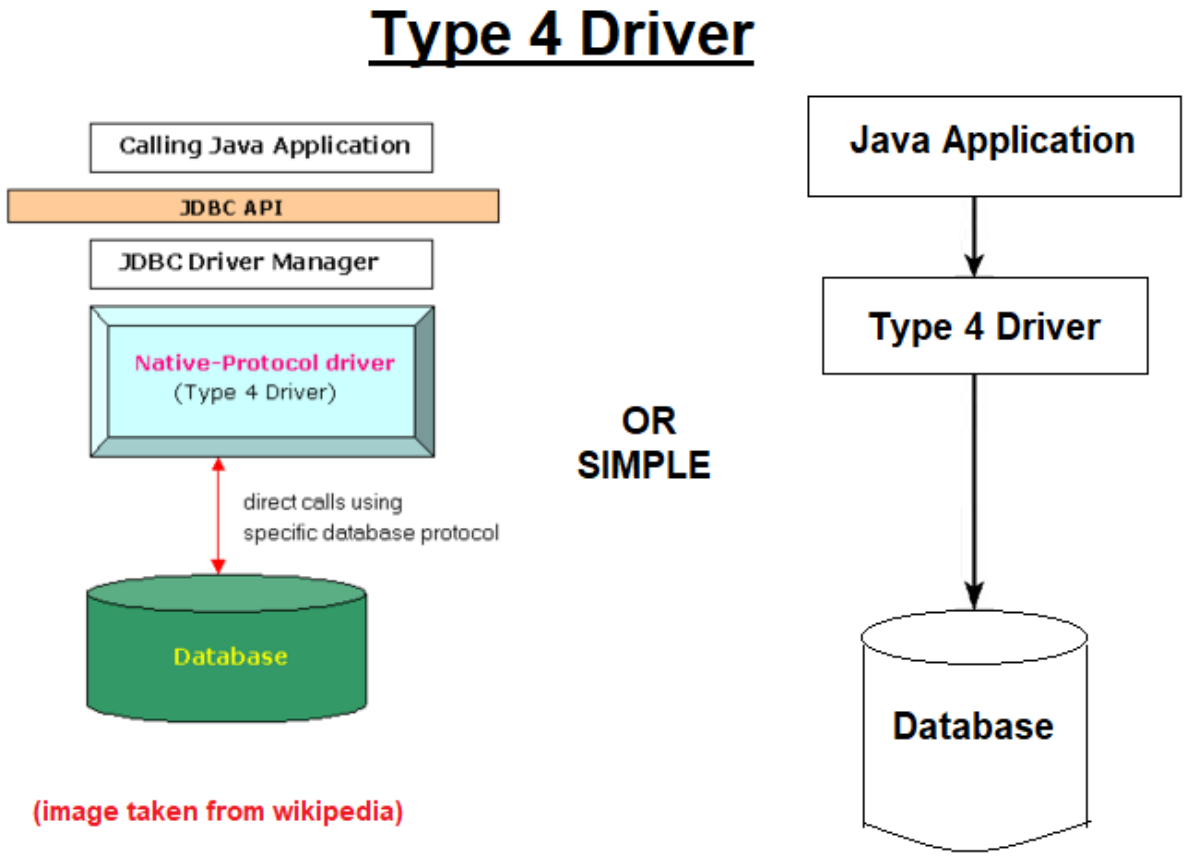
Password: root (MySQL) or system/tiger (Oracle)

Jar files: jdk13drv.jar

Smart Programming
We Educate
We Develop

Type 4 driver

Architecture Diagram:



(image taken from wikipedia)

We Develop

Other names: All Java Native Protocol Driver OR Pure Java Driver OR Thin Driver

Architecture type: 2 Tier Architecture (because there are two parts i.e. Client Machine and Database)

Thin or thick driver: Thin driver

Description:

- Type 4 driver uses database specific native protocols to communicate with database
- It converts JDBC calls into database specific calls directly
- This driver won't require any ODBC driver or native libraries at client side or Middleware server & hence it is known as thin driver
- This driver is developed only in Java & hence it is also known as pure java driver

Does it use any other driver: No

Other driver details: NA

Database dependency: Database Dependent because it communicates with the database directly

Platform dependency: Platform Independent as the driver is created in Java.

Used for: Used for both Standalone Applications & Web Applications or Distributed Applications or Enterprise Application

Advantages:

1. Completely implemented in Java to achieve platform independence.
2. Doesn't require to install any third party driver like ODBC or Native Libraries or Middleware Server
3. More security as it uses database vendor specific native protocol to communicate with database

Disadvantages:

1. The only limitation is Platform Dependent

Can be used with MySQL: Yes

Can be used with Oracle: Yes

Working:

Driver Class Name: com.mysql.jdbc.Driver (for MySQL)

oracle.jdbc.driver.OracleDriver OR oracle.jdbc.OracleDriver (for Oracle)

JDBC URL/Connection URL:

jdbc:mysql://localhost:3306/databasename (for MySQL)

OR jdbc:oracle:thin:@localhost:1521:XE

Username: root (MySQL) or system/scott (Oracle)

Password: root (MySQL) or system/tiger (Oracle)

Jar files: mysql-connector.jar (for MySQL) AND

ojdbc14.jar OR ojdbc6.jar OR ojdbc7.jar (for Oracle)

Smart Programming
We Educate
We Develop

Company Links & Contacts

Company Name: Smart Programming (+91 62838-30308)

Address : Chandigarh & Mohali (Punjab), India

Websites: https://www.smartprogramming.in/

https://courses.smartprogramming.in

Android App:
https://play.google.com/store/apps/details?id=com.smartprogramming

YouTube Channel:
https://www.youtube.com/c/SmartProgramming